Artificial Intelligence

Agent Title: **Construction Site Safety Monitoring Agent**

Textual Form

Sensor: Receives CCTV video, machinery proximity readings, safety sensor alerts, and site inspection reports.

Processing: Checks incoming information for unsafe conditions, such as missing safety helmets, workers too close to moving equipment, or blocked work areas. Evaluates risk and decides when to alert a supervisor.

Actuator: Sends real-time warnings to the site supervisor, displays hazard alerts on the dashboard, and records incidents for follow-up.

Formulation

Graphical Representation

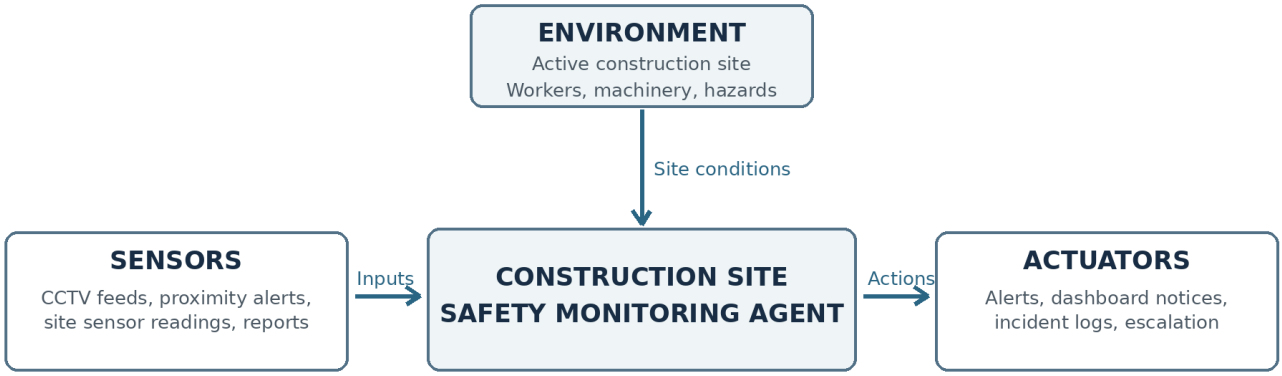


Figure: The agent observes site conditions and reports detected risks to responsible staff.

PEAS Framework

Performance → Detect hazards quickly and accurately; reduce missed hazards and unnecessary alerts.

Environment → Construction site, workers, supervisors, vehicles, machinery, materials, and changing weather.

Actuators → Issue safety alerts, notify supervisors, update the dashboard, and log incidents.

Sensors → CCTV cameras, proximity sensors, environmental readings, and inspection reports.

Environment Type

Property	Type	Why
Observability	Partially observable	Cameras and sensors cannot see every area or hidden danger.
Determinism	Stochastic	People, equipment, and site conditions can change unpredictably.
Episodes	Sequential	Earlier hazards and alerts affect later monitoring decisions.
Change	Dynamic	Workers, machines, and hazards move during construction work.
Values	Mixed	Distances and sensor values are continuous; alert categories are discrete.
Agents	Multi-agent	Workers, supervisors, drivers, and other systems interact on site.